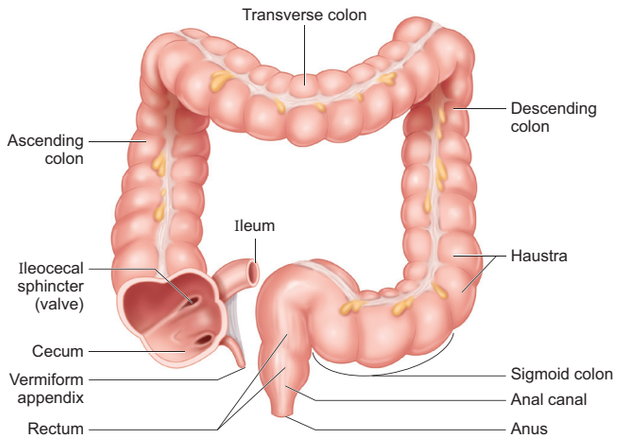


Figure 5.4 Large intestine

5.1.2 Histology of the Gut

The wall of the alimentary canal from oesophagus to rectum consists of four layers (Figure 5.5) namely **serosa, muscularis, sub-mucosa and mucosa.** The serosa (visceral peritoneum) is the outermost layer and is made up of thin squamous epithelium with some connective tissues. Muscularis is made of smooth circular and longitudinal muscle fibres with a network of nerve cells and parasympathetic nerve fibres which controls peristalsis. The submucosal layer is formed of loose connective tissue containing nerves, blood, lymph vessels and the sympathetic nerve fibres that control the secretions of intestinal juice. The innermost layer lining the lumen of the alimentary canal is the mucosa which secretes mucous.

5.1.3 Digestive glands

Digestive glands are exocrine glands which secrete biological catalysts called enzymes. The digestive glands associated with the alimentary canal are salivary glands, liver and pancreas. Stomach wall has gastric glands that secrete gastric juice and the intestinal mucosa secretes intestinal juice.

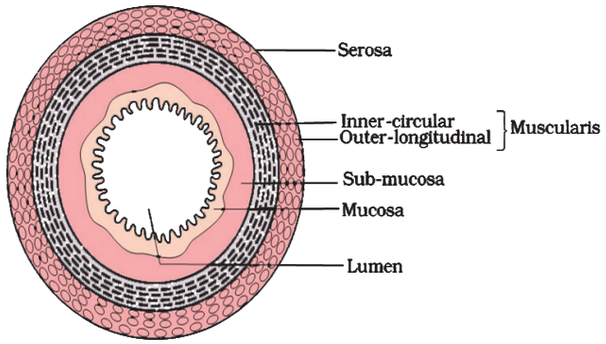


Figure 5.5 The layers of the alimentary canal

Salivary glands

There are three pairs of salivary glands in the mouth. They are the largest parotids gland in the cheeks, the sub-maxillary/ sub-mandibular in the lower jaw and the sublingual beneath the tongue. These glands have ducts such as **Stenson's duct, Wharton's duct and Bartholin's duct or duct of Rivinis** respectively (Figure. 5.6)**.** The salivary juice secreted by the salivary glands reaches the mouth through these ducts. The daily secretion of saliva from salivary glands ranges from 1000 to 1500mL.

Gastric glands

The wall of the stomach is lined by gastric glands. Chief cells or **peptic cells or zymogen cells** in the gastric glands secrete gastric enzymes and **Goblet cells** secrete mucus. The **Parietal or oxyntic cells** secrete HCl and an intrinsic factor responsible for the absorption of Vitamin B_{12} called **Castle's intrinsic factor**.

Liver

The liver, the largest gland in our body is situated in the upper right side of the abdominal cavity, just below the diaphragm. The liver consists of two major left and right lobes and two minor lobes. These lobes are connected with diaphragm. Each lobe has many hepatic lobules (functional unit of liver) and is covered by a thin connective

TN_GOVT_XI_Zoology _Ch05.indd 77 09-01-2020 5.14.59 PM